

TOOWOOMBA, Australia

WMO: 955510

Lat: 27.5425S

Lon: 151.9133E

Elev: 2105

StdP: 13.61

Time Zone: 10.00 (E10)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.3	40.7	20.0	16.2	59.8	23.7	19.3	56.3	26.8	53.1	24.6	51.9	10.2	240	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.8	90.3	66.0	87.3	66.0	84.4	65.6	71.9	80.5	70.6	79.0	69.4	77.5	12.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.5	117.5	74.5	68.3	112.3	73.4	67.2	108.0	72.4	37.0	80.5	35.9	79.0	34.9	77.4	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	23.8	22.0	32.9	96.2	2.1	3.5	31.3	98.7	30.1	100.7	28.9	102.6	27.3	105.2
			30.5	74.0	1.5	2.1	29.4	75.5	28.5	76.7	27.6	77.9	26.5	79.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	72.8	71.7	69.5	64.4	58.5	53.8	52.9	55.3	61.1	65.2	68.4	71.1
	DBStd	8.33	4.24	4.42	3.73	3.72	4.30	4.58	4.45	4.84	5.37	5.32	5.10	4.94
	HDD50	51	0	0	0	0	3	18	21	9	0	0	0	0
	HDD65	1509	1	2	4	52	203	336	374	304	141	63	22	7
	CDD50	5046	707	607	604	434	267	132	112	173	334	470	552	654
	CDD65	1028	243	189	143	35	3	0	0	3	24	68	124	196
	CDH74	6434	1617	1095	637	111	11	1	0	22	196	521	874	1349
	CDH80	1875	522	334	137	6	1	0	0	5	34	130	264	442
Wind	WSAvg	12.6	14.4	14.7	14.5	12.5	10.9	11.0	10.6	11.3	11.8	12.9	13.4	13.6
Precipitation	PrecAvg	36.9	4.8	4.6	3.1	2.3	2.7	1.9	1.9	1.6	1.5	3.0	3.9	5.0
	PrecMax	58.2	20.5	16.6	6.9	16.6	20.5	14.7	7.4	7.4	4.3	9.2	11.9	10.7
	PrecMin	18.5	0.7	0.4	0.1	0.0	0.2	0.1	0.0	0.0	0.1	0.3	0.6	0.4
	PrecStd	9.2	3.4	3.4	2.0	2.8	3.3	2.4	1.7	1.3	1.2	1.9	2.4	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.3	93.7	89.6	80.9	75.2	69.0	69.7	76.5	84.2	89.0	90.9	93.8
		MCWB	68.6	68.7	65.7	62.5	59.3	56.3	54.7	56.3	59.5	60.9	63.1	64.7
	2%	DB	89.0	88.1	83.9	77.2	71.3	65.9	66.1	71.0	79.2	83.5	86.6	89.0
		MCWB	67.5	68.4	66.7	62.3	57.7	55.5	53.4	53.8	57.6	60.1	63.4	65.6
	5%	DB	85.7	83.8	80.2	74.6	68.8	63.8	63.5	67.6	75.5	79.7	82.7	84.8
		MCWB	67.4	68.0	65.8	61.3	56.4	54.4	52.6	52.7	56.8	59.6	63.1	65.7
	10%	DB	82.3	80.4	77.4	72.2	66.6	61.6	61.1	64.4	71.9	76.1	79.0	81.2
		MCWB	67.3	67.1	65.1	60.7	55.8	53.5	51.7	51.6	56.0	59.1	62.8	65.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	73.9	71.7	67.1	63.6	60.6	61.1	60.3	63.5	67.5	69.9	72.0
		MCDB	83.0	84.5	78.3	74.4	67.5	64.3	64.6	69.4	72.5	77.1	81.4	83.1
	2%	WB	71.9	71.8	70.0	65.1	61.4	58.7	58.3	57.9	61.6	65.0	68.0	70.2
		MCDB	80.5	79.9	76.4	71.6	66.3	62.4	61.5	64.9	71.6	73.6	76.8	79.6
	5%	WB	70.7	70.2	68.5	63.8	59.9	57.2	55.9	55.9	60.0	63.4	66.5	69.0
		MCDB	78.9	78.5	74.4	69.8	64.7	60.7	59.9	62.8	69.7	71.8	75.4	77.5
	10%	WB	69.4	68.8	67.4	62.7	58.5	55.7	53.8	54.2	58.4	61.9	65.1	67.7
		MCDB	77.4	77.0	73.2	68.5	63.5	59.3	58.5	61.0	67.0	70.4	73.6	75.8
Mean Daily Temperature Range	5% DB	MDBR	16.8	15.6	15.1	15.3	15.4	14.4	15.8	17.7	18.8	18.7	18.1	17.3
		MCDBR	22.6	21.1	19.5	18.1	17.6	16.1	17.8	20.6	23.2	24.4	23.9	23.0
		MCWBR	6.3	6.0	5.9	6.3	7.5	7.9	8.6	9.1	8.0	8.0	7.3	7.0
	5% WB	MCDBR	17.5	17.5	15.0	14.4	13.0	11.8	12.7	16.1	19.0	18.6	18.8	18.2
MCWBR		6.8	6.6	6.2	6.1	6.4	7.0	8.1	8.7	7.9	7.8	7.4	7.1	
Clear-Sky Solar Irradiance	taub	0.351	0.343	0.324	0.307	0.288	0.282	0.275	0.285	0.318	0.339	0.349	0.358	
	taud	2.552	2.587	2.642	2.660	2.668	2.688	2.695	2.636	2.527	2.498	2.510	2.509	
	Ebn at Noon	314	312	309	299	292	288	295	303	305	309	313	312	
	Edh at Noon	35	33	29	26	23	22	23	26	32	35	36	36	
All-Sky Solar Radiation	RadAvg	2152	1968	1711	1484	1220	1010	1155	1456	1783	1989	2158	2159	
	RadStd	172	180	104	69	63	78	88	107	140	191	203	215	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.74	+0.94	N/A	N/A	N/A	N/A	N/A	-121	+335	+202

Nomenclature: See separate page